

About Nivakaran Shanmugabavan (Nivakaran S.)

Who I Am

Hi, I'm Nivakaran Shanmugabavan, a software engineer deeply passionate about bridging the world of data science, artificial intelligence, and practical engineering. My work revolves around building smart, scalable, and purposeful AI systems, from real-time computer vision apps to language models and agentic assistants. I am currently a B.Sc(Hons) in Information Technology undergraduate at SLIIT.

I love teaching machines to understand the world, whether that's through natural language, images, or structured data, and designing systems that feel human, intuitive, and intelligent.

My Projects

Main Projects

Project 01: EcoHarvest

Description: A sustainable food resale platform built with MongoDB, Express.js, Next.js, and Node.js, designed to reduce food waste by connecting businesses with surplus food to budget-conscious consumers. Features real-time inventory tracking and dynamic pricing based on expiry dates.

Problem: Addresses food waste due to oversupply, expiry dates, and supply chain inefficiencies.

Solution: Connects surplus food sources with consumers and recycling partners, ensuring quality checks and optimized logistics.

Tech Stack: MongoDB, Express.js, Node.js, Next.js.

Project 02: Professional Portfolio

Description: A dynamic full-stack portfolio platform built with MongoDB, Express.js, Next.js, and Node.js. Features an admin dashboard for blog updates and a responsive UI.

Problem: Static portfolios lack dynamic features and require manual updates.

Solution: A CMS-enabled portfolio with SEO optimization and GitHub integration.

Tech Stack: Next.js, Node.js, MongoDB, Express.js.

Project 03: Finance Tracker Mobile App

Description: An Android app built with Kotlin to track expenses, categorize spending, and set budgets with real-time alerts.

Problem: Users struggle to manage finances manually.

Solution: Offline-first app with interactive dashboards and budget alerts.

Tech Stack: Kotlin.

Project 04: Book Recommendation System

Description: Uses NLP and semantic search to recommend books based on user input (description, category, emotional tone).

Problem: Traditional systems fail to interpret nuanced queries.

Solution: Employs Hugging Face models and ChromaDB for semantic understanding.

Tech Stack: Python, Hugging Face, Chroma, Gradio.

Project 05: Movie Recommendation System

Description: Recommends movies based on metadata (title, overview, cast) using cosine similarity and TMDB API.

Problem: Generic recommendations lack personalization.

Solution: Content-based filtering for tailored suggestions.

Tech Stack: Python, Streamlit.

Project 06: Fraud Detection in Transactions

Description: ML pipeline to detect fraudulent transactions with MLflow tracking.

Problem: Manual fraud detection is error-prone.

Solution: Automated pipeline with real-time predictions.

Tech Stack: Python, Sklearn, MLflow.

Project 07: Delivery Time Prediction

Description: Predicts delivery times using traffic, weather, and personnel data.

Problem: Static estimates are inaccurate.

Solution: Real-time ML model with Flask integration.

Tech Stack: Python, MLflow, Sklearn.

Project 08: AI Sign Language Interpreter

Description: Real-time hand gesture recognition using OpenCV and MediaPipe.

Problem: Traditional input devices are inaccessible.

Solution: Customizable gesture system for touchless interaction.

Tech Stack: Python, OpenCV, MediaPipe.

Project 09: Food Image Classification

Description: Classifies food images into 101 categories using EfficientNet-B0.

Problem: Manual labeling is slow and costly.

Solution: Offline-capable model with FastAPI backend.

Tech Stack: Python, PyTorch, FastAPI.

Project 10: Max: Portfolio Chatbot

Description: AI chatbot answering questions about skills/projects using Groq's Llama model.

Problem: Static portfolios lack interactivity.

Solution: Context-aware chatbot with session history.

Tech Stack: FastAPI, LangChain, Groq.

Project 11: NarrateX: Blog Generation AI

Description: Multilingual blog generator (English, Tamil, Sinhala) using Groq and LangGraph.

Problem: Limited AI support for non-English languages.

Solution: Modular workflow for dynamic content creation.

Tech Stack: Python, LangGraph, Groq.

Project 12: Network Security Data Pipeline

Description: ML pipeline for threat detection with MongoDB and MLflow.

Problem: Manual monitoring is inefficient.

Solution: Automated anomaly detection.

Tech Stack: Python, MLflow.

Project 13: Custom Virtual Background Broadcaster

Description: Real-time background replacement using YOLOv8 and OpenCV.

Problem: Existing tools lack customization.

Solution: Cross-platform solution with FastAPI.

Tech Stack: Python, YOLO, OpenCV.

Project 14: Agentic AI Assistant

Description: Streamlit app for conversational AI, web search, and news summaries.

Problem: Static AI tools lack real-time capabilities.

Solution: Multi-tool integration with LangGraph.

Tech Stack: Python, LangGraph, LangChain.

Mini Projects

1. Search Agent: Web/research assistant using LangChain and Groq.
2. Conversational RAG Q&A: Chat with PDFs using HuggingFace embeddings.
3. SQL Database Chatbot: Query databases in plain English.
4. YouTube & Website Summarizer: 300-word summaries using LangChain.
5. Math Problem Solver: Solves problems with stepwise explanations.
6. Malaria Detection Web App: CNN model for blood smear analysis.
7. Cat/Dog/Person Classifier: Gradio app with PyTorch.
8. IMDB Sentiment Analyzer: Classifies movie reviews.

Case Studies

Case Study 01: Professional Portfolio

Challenge: Dynamic content management, multi-domain organization, security.

Solution: Tiptap CMS, JWT authentication, AI chatbot (Max).

Results: Enhanced engagement, scalable updates.

Case Study 02: EcoHarvest

Challenge: Food waste, logistics, consumer trust.

Solution: Real-time inventory, quality checks, partnerships.

Results: Operational platform with scalability.

Let's Connect

Whether you're a recruiter, collaborator, or just curious, I'd love to talk.

Find me at:

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